Can You Build a Model That Detects Fake News?

A DS 4002 case study in text classification, model evaluation, and responsible interpretation

Your Role	Your Mission
You are a junior data analyst working with a media literacy team. The team wants a simple, explainable model that can sort news articles into real or fake categories using article text and headlines.	Use the provided fake/real news dataset and starter project materials to explore the data, build a baseline text classifier, evaluate its performance, and explain what the model can and cannot tell us.

The Challenge
Fake news spreads quickly because it often looks and sounds like real reporting. A machine learning model can help flag suspicious content, but the model has to learn from patterns in text, not from true understanding. That makes this case study a chance to practice the full data science workflow while also thinking critically about what a model's predictions mean.

What You Will Work With
You will receive a GitHub repository containing Python scripts, cleaned data outputs, exploratory graphics, and a logistic regression model using TF-IDF text features. You are not expected to invent a perfect fake news detector. Your goal is to understand the project, reproduce the workflow, and communicate the model's strengths and limitations clearly.

Your Final Deliverable
Produce a concise case-study report or notebook that teaches another student how a fake news classifier works. Your final product should explain the project motivation, walk through the general modeling process, summarize the model's results, and reflect on the strengths and limitations of automated fake news detection.

Why This Matters
This project is not just about getting a high accuracy score. It is about learning how text becomes data, how baseline models make predictions, and how to explain a technical result to a non-expert audience. By the end, you should be able to say not only whether the classifier worked, but also why it worked, when it might fail, and how you would improve it.

Big Question: How can we build a simple model to classify real vs. fake news while staying honest about the limits of automated misinformation detection?

To get started visit https://github.com/zane-abbud/DS4002-Case-Study